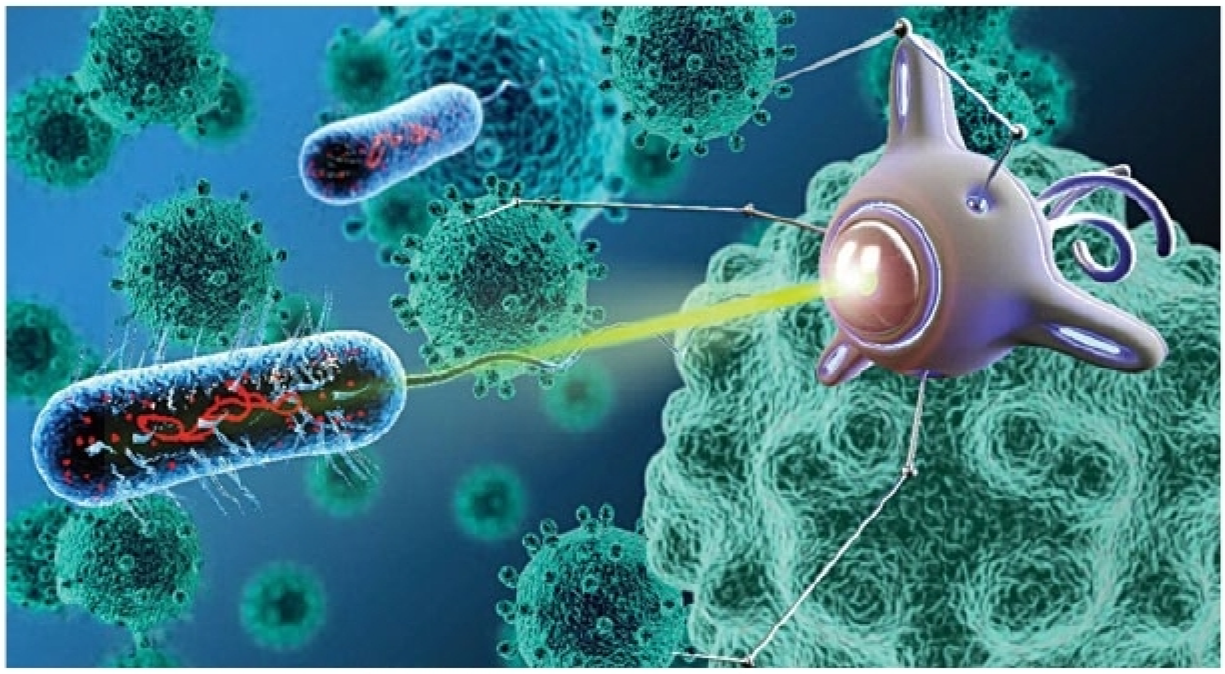

Bacteriobot, a medical nanorobot used to treat cancer (Credit: http://longevityfacts.com)

minimised surgical procedures. These have the ability to enter the bloodstream, be controlled wirelessly and applied anywhere in the body for treatment of sensitive body parts such as the eye.

With micro-robots propelled in the blood circulatory system, a large number of remote locations in the human body become accessible. With blood vessels diameters varying from 25mm as aorta to 0.01mm as capillaries, propelling micro-devices wirelessly and their locomotion at the microscopic level becomes a challenge. The robots' shape is programmed for smooth travel through fluids that are dense, viscous or moving at rapid speeds to enable them to become accustomed to the characteristics of the fluid they are travelling through.

For example, when there is a change in viscosity or osmotic concentration, they swiftly alter their shape to maintain speed and maneuverability, without losing control of the direction of motion, mimicking the ability of naturally-occurring micro-organisms to adjust their shape. The natural ability is mimicked by using a controlling electromagnetic field to automatically transform them into the most efficient shape to maximise performance.

Another technique is to use rigid chains of micro- and nano-particles, which produce propulsion when rotated by a magnetic field. On actuation, the micro-swimmers display a spiral motion, resulting in forward propulsion.

Other uses in the medical field include stem cell delivery to a particular region of the body; monitoring of chemical and physical parameters inside the body; and addressing pathways inside the human body like gastrointestinal tract, blood circulatory system or central nervous system. Smart endoscopic pills travel through the gastrointestinal tract and provide real-time feedback from inside the body.

Another useful application is support in repair of tissue cells. Micro-bots attach themselves to the surface of recruited white cells and squeeze their way out through the walls of blood vessels arriving at the injury site, where they assist in the tissue repair process.

Micro-bots for fighting cancer. Scientists from Chonnam National University, South Korea, have developed Bacteriobot, which is capable of fighting cancers. Genetically-modified, non-toxic bacteria are attached to micro-bots with a bead to precisely attack tumour cells in the human body. When injected into the blood stream, they diagnose and treat cancer by migrating and targeting tumours, deliver the drug directly to the tumours and attack the tumours, leaving the healthy cells alone. The patient is spared the side-effects of chemotherapy.

Limitation of Bacteriobot is its capability to detect only solid tumour-forming cancers such as breast cancer and colorectal cancer.

Micro-bots in surgery. One of the most delicate organs in human body is the eye. Multi Scale Robotics Lab, Zurich, has developed a magnetically-